

A High-Efficiency and Incentive-Compatible Peer-to-Peer Energy Trading Mechanism

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Abstract—Due to the rapid development of demand response management and distributed energy resources, prosumers are becoming more proactive, which also promotes the emergence of peer-to-peer (P2P) energy trading mechanisms. However, we find that it is quite difficult to simultaneously achieve high computational efficiency, decentralized operations and solution optimality in a P2P energy trading mechanism, which is called the “P2P energy trading trilemma”. In this paper, we first propose a novel negotiation mechanism for P2P energy trading that can maximize social welfare in a decentralized manner and respect physical network constraints. Then, the local optimization problem is transformed into a closed-form alternating update algorithm (AUA), so that the computational efficiency can be greatly improved. Therefore, our proposed mechanism solves the “P2P energy trading trilemma” to a certain extent. Another challenge is that the locational marginal price (LMP)-based market mechanism cannot satisfy the incentive-compatible property. To encourage prosumers to cooperate in a lack-of-trust environment, a novel incentive-compatible mechanism is proposed for P2P energy trading using consensus mechanism inspired by proof of solution (PoSo) and smart contract (SC). Finally, we simulate the functionality of the mechanism in terms of convergence performance, reliability, scalability, computational efficiency, and SC operations.

Index Terms—P2P energy trading trilemma, consensus alternating direction method of multipliers, power transfer

Manuscript received 18 October 2022; revised 17 January 2023 and 6 March 2023; accepted 8 April 2023. Date of publication 13 April 2023; date of current version 26 December 2023. This work was supported in part by the Beijing Natural Science Foundation under Project M21031; in part by the Natural Science Foundation of China under Project U21A20467, Project 61932011, and Project 61972019; in part by the National Key R&D Program of China under Grant 2022YFB2701600; in part by the Zhejiang Soft Science Research Program under Grant 2023C35081; in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LZ23F020012; and in part by the Open Research Fund of Key Laboratory of Cryptography of Zhejiang Province under Grant ZCL21007. Paper no. TSG-01562-2022. (Corresponding author: Bo Qin.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TSG.2023.3266809>.

Digital Object Identifier 10.1109/TSG.2023.3266809

distribution factor, alternating update algorithm, proof of solution, smart contract.

I. INTRODUCTION

THE GROWING number of prosumers naturally necessitates a decentralized energy trading mechanism that allows prosumers to trade freely with each other. With the advent of modern technologies, a fully decentralized network architecture, which can be defined as a peer-to-peer (P2P) network, is emerging for participants in the market to share a portion of their own resources with each other. The formal definition of P2P networks can be found in [1]. In this context, P2P trading mechanisms have emerged as the next generation of energy management techniques that enable prosumers to actively participate in the energy market. The use of P2P energy trading mechanisms can provide numerous benefits, including reducing peak demand [2], minimizing reserve requirements [3], and increasing power system resilience [4]. It is worth noting that, electricity exchange is quite different from any other exchange of goods. This is due to the fact that peers are part of a power network, which has its own physical technical constraints on energy exchange [5].

Different decentralized methods are employed to realize P2P energy trading mechanism. A primal-dual gradient algorithm [6] is proposed to clear the market in a fully decentralized manner, and [7] proposed modified primal-dual sub-gradient algorithm (PDSGA); relaxed consensus+innovation (RCI) [8] is designed to solve the multi-bilateral economic dispatch in fully decentralized manner. Each market participant aims at solving his own local problem while reacting on the primal and dual estimates of others; bilateral contract networks [9] is proposed as a new scalable market design for peer-to-peer energy trading; double action methods [10], [11] seek energy matching pairs from the supply and demand sides, and encouraging direct energy trading between producers and consumers; an iterative peer matching process is employed to match peers for bilateral negotiation [12]; a game-theoretic model is proposed for real-time P2P energy trading in a prosumer-based community microgrid [13], an optimal pricing strategy using Stackelberg game for credit-based loans in P2P energy trading is proposed [14], and two noncooperative demand-response games [15] are analyzed in P2P energy trading; standard alternating direction method of multipliers (ADMM) [16], [17] is an efficient tool for P2P energy trading but a central market operator is needed to help with convergence;

to build a fully decentralized market, consensus ADMM (CADMM) [18], [19], [20], [21], [22] is proposed, which has good convergence performance and decentralized structure.

However, we observe that it is quite difficult to simultaneously achieve high computational efficiency, decentralized operation, and solution optimality in a P2P energy trading mechanism, which is first proposed in this paper as a “P2P energy trading trilemma”. To be specific, distributed optimization (DO)-based methods can operate in a decentralized manner and ensure the optimality of the solution, but suffer from the high computational burden; bilateral matching methods are computationally efficient, but still require a central coordinator and have difficulty guaranteeing optimality because agents have no flexibility to adjust bidding prices/quantities according to each other’s decisions. Centralized methods, e.g., interior point method (IPM), can guarantee optimal solutions, but scalability is limited and comes at the expense of decentralization.

Another challenge is to encourage prosumers to cooperate in such a trustless environment. In P2P energy trading, prosumers are expected to trade their energy among themselves without any influence of a central coordinator, making P2P platforms a trustless and unreliable system. Moreover, the most widely used locational marginal price (LMP)-based market mechanism cannot satisfy the properties of “incentive compatibility”¹ and “market efficiency,”² since a market player can exercise “market power” by not trustfully offering in terms of price and quantity. To address this challenge, blockchain has emerged as a promising, user-friendly and efficient technology for the implementation of secure and reliable P2P energy trading mechanisms. Existing studies have exploited a wide variation of blockchain-enabled platforms to ensure secure and transparent P2P energy trading [11], [14], [15], [18], [23], [24], [25], [26], [27], [28], [29]. It makes communication transparent to prosumers and allows them to make decisions about energy dispatches in a decentralized and untrusted environment. Consensus mechanisms, e.g., Proof of Work (PoW) and Proof of Stake (PoS), are efficient tools in the blockchain to verify the correctness and authenticity of submitted results. Smart contract (SC) also play a very important role in P2P energy trading, controlling the energy transactions between two peers by following predefined trading rules and preventing any single party from dominating and manipulating the outcome.

In this work, we focus on using the DO algorithms, which are widely adopted to design P2P negotiation mechanisms for energy trading. However, it is usually complex and requires specialized solvers to solve an optimization problem, which may consumes a large amount of computing resources. This also poses challenges on existing blockchain technologies that are not specialized for mathematical optimization. Some works [28], [30], [31] explored the combination of optimization approach with blockchain and SC in energy trading. Reference [28] proposed a P2P energy trading framework enabled by blockchain. A double-auction

Vickrey-Clarke-Groves (VCG) mechanism is processed on blockchain, but the optimization tasks are still solved by professional optimizer, which probably costs high computational burden; [30] proposed an alternative blockchain mechanism capable of searching for good approximate solutions to an optimization problem, which sacrificed exactness; [31] proposed a blockchain consensus mechanism PoSo tailored to support mathematical optimization problems. A delegation is selected for solving optimization problems and verification, however, the decentralization and democracy has been sacrificed to some extent. These works demonstrated that the “P2P energy trading trilemma” still exists in the area of blockchain-based P2P energy trading.

To overcome the troublesome trilemma, in this paper, we innovatively propose an efficient P2P energy trading mechanism combining DO method with SC, and design a consensus mechanism inspired by proof of solution (PoSo) [31] to build an incentive-compatible mechanism for P2P energy trading. Compared with our previous and others’ work, the contributions mainly lie as follows.

- 1) A novel P2P negotiation algorithm combining consensus ADMM and Power Transfer Distribution Factor (PTDF) is proposed for energy trading. This mechanism can maximize social welfare while maintaining the line flow constraints. Agents are guided not to transfer power through congested lines by introducing the line prices. Moreover, the line flows calculation and prices update is performed by the line manager, who behaves as a single agent, which does not break the decentralization of the market. Our proposed P2P energy trading mechanism can ensure the property of optimality and decentralization.
- 2) Although the CADMM method can reduce the iterations required for convergence, the computational cost of the local optimization problem is still high and cannot be executed by a SC in the blockchain. To overcome this drawback, we improve the algorithm by equivalently converting the local optimization problem into a closed-form alternating update algorithm (AUA), which can greatly improve computational efficiency. Thus, our designed algorithm can simultaneously have fewer iterations (merit of ADMM) and high computational speed (merit of closed-form AUA). Compared to existing work on P2P energy trading using DO methods, we increase the computational efficiency and implement the DO-based P2P energy trading mechanism through secure SC technology.
- 3) A consensus mechanism inspired by PoSo is proposed for P2P energy trading. Verifying a given solution satisfies the optimality Karush-Kuhn-Tucker (KKT) conditions is much easier than finding it. Compared to the original work [31], the main difference is that we do not require delegation, and agents do not need to pass all information to a central delegation to solve the optimization problem. Agents solve individual problems in parallel, and send the accumulated KKT conditions to neighboring agents for verification by deployed SC. Our consensus mechanism can overcome the shortcomings of the LMP-based mechanism and build an

¹A mechanism is called incentive-compatible if every participant can maximize its objective just by acting according to its true preferences.

²Market efficiency is maximized when outcomes maximize social welfare.

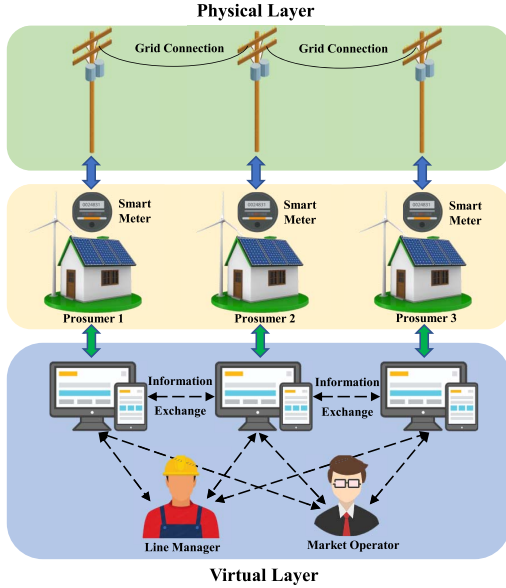


Fig. 1. An illustration of the two-layer P2P power network.

incentive-compatible market by making use of the consensus mechanism.

The rest of the paper is organized as follows: Section II presents the formulation of the P2P energy trading and social welfare maximization problem. Section III proposes the P2P negotiation and consensus mechanisms for P2P energy trading, and the implementation by SCs. Numerical results are presented in Section IV. Finally, conclusions, limitations and future perspectives are drawn in Section V.

II. PROBLEM FORMULATION OF P2P ENERGY TRADING

In this paper, a P2P power network is considered with a number of agents, who can act as sellers when they have energy production surplus from their renewable resources, or as buyers to satisfy their energy demand. The energy trading mechanism proposed in the following is for active power trading, and the emphasis is placed on a deterministic clearing algorithm for a day-ahead single time period, which can be easily extended to multiple time slots [22]. Without loss of generality, we assume that the length of time period is 1 hour, which means the values of energy and power are same. In this section, we first model the two-layer P2P power network, under which, the models of participants and physical line constraints is formulated. Finally, the social welfare maximization problem is proposed.

A. Blockchain-Based Two-Layer P2P Power Network

The network is divided into physical layer and virtual layer as shown in Fig. 1. The physical layer includes electrical connection of all agents. The physical layer is modeled as an undirected radial graph $\mathcal{G}(\mathcal{N}, \mathcal{L})$, consisting of a set $\mathcal{N}$ of nodes (buses) and a set $\mathcal{L}$ of distribution edges (lines) connecting these nodes. We index the buses in $\mathcal{N}$ by i , where bus 0 represents the root bus. We index the lines in $\mathcal{L}$ by l , also denote a line in $\mathcal{L}$ by the pair (i, j) of buses it connects.

We call j the parent of i , and the set of parents of bus i is denoted by $\pi(i)$. The physical network is required to be connected, which means each pair of agents in the network can trade and transfer power.

The virtual layer is the communication network used for information exchange among agents. In this work, the communication network is built based on consortium blockchain, and we consider the identities of these nodes/agents to be public, e.g., certified by a PKI, so that the trusted identity information is guaranteed. In the virtual layer, there is a set Ω of agents, who can be sellers or buyers at each time period, denoted by Ω_S and Ω_B , respectively. The set of sellers and buyers is indexed by n and m . Thus a communication link between a pair of seller and buyer can be denoted by (n, m) . We assume that the communication network is predetermined, and are forever preserved as fixed communication constraints. We assumed that all agents can communicate with each other and negotiate. At each line, there is a line manager (LM) who is in charge of computing power flows and updating line prices. Finally, market operator (MO) is responsible for checking if the negotiation process is completed and storing transaction results on the blockchain.

B. Models of Participants

At the beginning of each time period, each agent will be aware of its role, e.g., seller or buyer, depending on their predicted renewable energy production and demand. Sellers provide their surplus energy to buyers, and the cost of providing energy offered by each seller $\forall n \in \Omega_S$ can be modeled by a quadratic convex function [32] as below

$$C_n(E_n) = a_n E_n^2 + b_n E_n + c_n, \quad (1)$$

where a_n , b_n and c_n are predetermined positive constants. The power injection E_n of each seller $n \in \Omega_S$ is the sum of bilaterally traded quantities with buyers $m \in \Omega_B$ as

$$E_n = \sum_{m \in \Omega_B} E_{nm}, \quad \forall n \in \Omega_S. \quad (2)$$

To lighten notations, $\mathbf{E}_n = \{E_{n1}, \dots, E_{nm}\}$ is used to represent the whole set of transactions of seller n . The power set-points of a seller n is constrained by the boundaries as

$$\underline{E}_n \leq E_n \leq \overline{E}_n, \quad \forall n \in \Omega_S, \quad (3)$$

and the trading energy with buyer m is constrained as

$$0 \leq E_{nm} \leq E_n, \quad \forall (n, m) \in (\Omega_S, \Omega_B). \quad (4)$$

Buyers are equipped with flexible loads that allow them to adjust demand according to energy prices. Buyers' satisfaction is based on their demand and can be modeled by utility functions. The general form of utility function is a non-decreasing function with a non-increasing marginal benefit. The utility function for each buyer $\forall n \in \Omega_b$ with demand should can be modeled as [33]:

$$U_m(E_m) = \begin{cases} \varpi_m E_m - \varrho_m E_m^2, & E_m < \varpi_m / 2\varrho_m \\ \varpi_m^2 / 2\varrho_m, & E_m \geq \varpi_m / 2\varrho_m \end{cases} \quad (5)$$

where ϖ_m and ϱ_m are predetermined positive constants. Similarly, the net power injection E_m of each buyer is the sum of bilaterally traded quantities with sellers $n \in \Omega_S$ as

$$E_m = \sum_{n \in \Omega_S} E_{mn}, \quad \forall m \in \Omega_B, \quad (6)$$

and $\mathbf{E}_m = \{E_{m1}, \dots, E_{mn}\}$ is used to represent the whole set of transactions of buyer m . The power set-points of a buyer m is constrained by the boundaries as

$$\underline{E}_m \leq E_m \leq \overline{E}_m, \quad \forall m \in \Omega_B. \quad (7)$$

Here, we assume that maximal demand of buyer will not reach his satisfaction upper bound, i.e., $\overline{E}_m \leq \varpi_m/2\varrho_m$, since excess energy consumption does not bring more utility. The trading energy with seller n is constrained as

$$0 \leq E_{mn} \leq E_m, \quad \forall (n, m) \in (\Omega_S, \Omega_B). \quad (8)$$

Above similar models have also been proposed in our previous works [21], [22].

C. Line Flow Constraints of Power Network

In this work, PTDF is used to calculate line flows and indicate lines that are used for power transfer in each individual transaction [6]. PTDF for line l is denoted by φ_{ij}^l and indicates portion of the generated energy by agents at bus i which is transferred to agents at bus j through line l . φ_{ij}^l can be calculated by $\varphi_{ij}^l = \psi_i^l - \psi_j^l$, where ψ_i^l, ψ_j^l are Injection Shift Factors (ISF) in line l for bus i and j . The ISF matrix is $\Psi \triangleq [\psi_i^l] \in \mathbb{R}^{L \times N}$, where N is the number of buses in the network and L is the number of lines. This matrix can be computed as $\Psi \triangleq B'AC^{-1}$ by having diagonal branch susceptance matrix B' , branch to node incidence matrix A , and reduced nodal susceptance matrix C . In the matrix A , a_l^T is l^{th} row, in which a line exist connecting bus i and j .

$$B' \triangleq \text{diag}[b_1, b_2, \dots, b_L] \in \mathbb{R}^{L \times L} \quad (9a)$$

$$A \triangleq [a_1, a_2, \dots, a_L] \in \mathbb{R}^{L \times N},$$

$$a_l^T \triangleq \begin{bmatrix} 0 & \dots & 0 & i & 0 & \dots & 0 & j & 0 & \dots & 0 \\ & & & 1 & & & & -1 & & & \end{bmatrix} \quad (9b)$$

$$C \triangleq A^T B' A \in \mathbb{R}^{N \times N} \quad (9c)$$

By having PTDF matrix and traded energy between agents, the power flow in line l can be obtained by (10)

$$P_l = \sum_{n \in \Omega_S} \sum_{m \in \Omega_B} \varphi_{ij}^l E_{nm}. \quad (10)$$

and seller n locates at bus i and buyer m locates at bus j . Their traded power E_{nm} has an impact on the flow P_l . If the value of P_l is lower or higher than the boundaries, price signals $(\overline{v}_l, \underline{v}_l)$ will be sent to agents who are using the particular line to transfer power, in order to avoid overflow and congestion.

In order to avoid damage to the transmission lines, the real power flow P_l in each line l is bounded by maximum capacity $P_l^{\max}$ related to the heat that they can dissipate

$$-P_l^{\max} \leq P_l \leq P_l^{\max}, \quad \forall l \in \mathcal{L}. \quad (11)$$

Remark 1: P2P energy trading is usually proposed in the distribution network, where ACOPF is often considered. To

consider ACOPF, network branch flow can be described using the LinDistFlow [34] model, and a system operator (SO) is introduced into the market to help complete the calculation of power flows and bus voltage magnitudes by solving a convex optimization problem transformed from the LinDistFlow model. In this work, the PTDF model is preferred to utilize the function of line prices, and we believe that our proposed method can be used and workable in some high-voltage distribution networks to increase computation efficiency.

D. Social Welfare Maximization Problem

The equilibrium between sellers and buyers is represented by balance constraints, and in P2P network, they can be replaced by a set of reciprocity constraints formulated as

$$E_{nm} = E_{mn}, \quad \forall (n, m) \in (\Omega_S, \Omega_B). \quad (12)$$

From above, the P2P energy trading has the goal to maximize social welfare under the constraints. Since the cost functions are strictly convex and utility functions are strictly concave, the optimal point will exist and be unique [35]. Mathematically, the problem of maximizing social welfare can be formulated as

$$\max \sum_{m \in \Omega_B} U_m(E_m) - \sum_{n \in \Omega_S} C_n(E_n) \quad (13a)$$

$$\text{s.t. } (2) - (4), (6) - (8), (11) \text{ and } (12) \quad (13b)$$

Since the above problem is a convex optimization problem, the optimal solutions can be computed by a wealth of centralized methods. However, it does not protect privacy because it requires the disclosure of all information about the agents. To this end, a fully decentralized negotiation mechanism is ideal. Next, we will design a P2P negotiation mechanism for energy trading that can realize the optimal scheduling of the problem (13).

III. NEGOTIATION AND CONSENSUS MECHANISM FOR P2P ENERGY TRADING

In this section, we proposed a novel decentralized negotiation mechanism that combines consensus ADMM, PTDF, and AUA. The line price is incorporated into the local objective function to guide agents to spontaneously adjust the power to satisfy the line flow constraints. To further improve the computational efficiency, we design the closed-form AUA based on KKT conditions. We also propose a consensus mechanism for P2P energy trading based on PoSo mechanism. Finally, we implement P2P energy trading through secure SC. An illustration of the blockchain-enabled P2P energy trading architecture is shown in Fig. 2.

A. Local Update of Agents

The decentralized negotiation algorithm for P2P energy trading is detailed as follows. Initially, all agents will initiate a pair of price and amount of energy in parallel, and send to neighbors. Then each one will update their energy quantities and prices for neighbors.

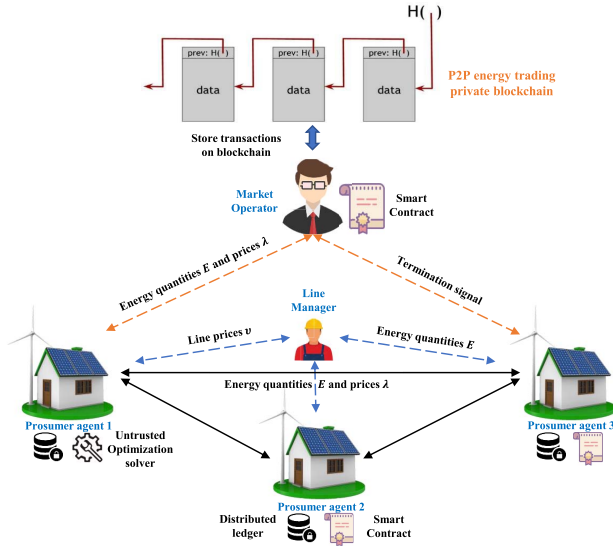


Fig. 2. An illustration of blockchain-enabled P2P energy trading architecture.

For seller: Each seller n in bus i will update its energy trading amount to buyers by solving a local optimization problem at a given iteration k as shown below.

$$\begin{aligned} \min_{\mathbf{E}_n^{k+1}} \quad & C_n(E_n) + \sum_{l \in \mathcal{L}} \sum_{m \in \Omega_B} \phi_{ij}^l (\bar{v}_l^k - \underline{v}_l^k) E_{nm} \\ & + \sum_{m \in \Omega_B} \left[\lambda_{nm}^k (F_{nm}^k - E_{nm}) + \frac{\rho}{2} (F_{nm}^k - E_{nm})^2 \right] \\ \text{s.t.} \quad & (2) - (4). \end{aligned} \quad (14)$$

where ρ is positive penalty factor; λ_{nm} is the dual variable of the balance constraints (12), also denote the price provided by seller n for the traded energy power E_{nm} to buyer m ; $F_{nm}^k = \frac{E_{nm}^k + E_{mn}^k}{2}$ represents the consensus value.

Sellers will update the energy prices λ_{nm}^{k+1} provided for buyers as follows

$$\lambda_{nm}^{k+1} = \left[\lambda_{nm}^k - \frac{\rho}{2} (E_{nm}^{k+1} - E_{mn}^{k+1}) \right]^+, \quad (15)$$

where $[\cdot]^+$ denotes the $\max(\cdot, 0)$.

For buyer: Similarly, each buyer m in bus j will update its demanded energy to sellers by solving the local optimization problem at iteration k as below.

$$\begin{aligned} \min_{\mathbf{E}_m^{k+1}} \quad & -U_m(E_m) - \sum_{l \in \mathcal{L}} \sum_{n \in \Omega_S} \phi_{ji}^l (\bar{v}_l^k - \underline{v}_l^k) E_{mn} \\ & + \sum_{n \in \Omega_S} \left[-\lambda_{nm}^k (F_{mn}^k - E_{mn}) + \frac{\rho}{2} (F_{mn}^k - E_{mn})^2 \right] \\ \text{s.t.} \quad & (6) - (8). \end{aligned} \quad (16)$$

Here, $F_{mn}^k = \frac{E_{mn}^k + E_{nm}^k}{2}$, and it is noted that $F_{mn}^k = F_{nm}^k$.

For LM: All agents will also broadcast their traded energy $\mathbf{E}^{k+1}$ to LM, who is responsible for updating the power flows of each line and checking if the power flow is within the security boundary. LM will update the power flow of each line by (10), and update the two Lagrangian multipliers which

act as line price signals $(\bar{v}_l^{k+1}, \underline{v}_l^{k+1})$ as below

$$\bar{v}_l^{k+1} = \left[\bar{v}_l^k + \phi(P_l^{k+1} - P_l^{\max}) \right]^+, \quad (17a)$$

$$\underline{v}_l^{k+1} = \left[\underline{v}_l^k - \phi(P_l^{k+1} + P_l^{\max}) \right]^+. \quad (17b)$$

LM will send line price signals to the corresponding agents who are using the particular line for energy trading. To be specific, from (14), the final line utilization on a trade from seller n to buyer m , who are located at bus i and j respectively, is $\sum_{l \in \mathcal{L}} \phi_{ij}^l (\bar{v}_l - \underline{v}_l)$. By applying price signals to line utilization, agents will try not to trade energy using congested line.

For MO: Since $C_n(E_n)$ and $-U_m(E_m)$ are closed, proper and convex, it is sufficient to ensure convergence to the global optimal point of our algorithm [36]. The algorithm converges as long as the total local primary and dual residuals fall below global stopping criteria,

$$S^{k+1} \triangleq \sum_{n \in \Omega_S} \sum_{m \in \Omega_B} (E_{nm}^{k+1} - E_{mn}^{k+1})^2 \leq \chi^s, \quad (18a)$$

$$T^{k+1} \triangleq \sum_{n \in \Omega_S} \sum_{m \in \Omega_B} (E_{nm}^{k+1} - E_{nm}^k)^2 \leq \chi^t, \quad (18b)$$

$$U^{k+1} \triangleq \sum_{n \in \Omega_S} \sum_{m \in \Omega_B} (\lambda_{nm}^{k+1} - \lambda_{nm}^k)^2 \leq \chi^u, \quad (18c)$$

$$V^{k+1} \triangleq \sum_{l \in \mathcal{L}} \left[(\bar{v}_l^{k+1} - \bar{v}_l^k)^2 + (\underline{v}_l^{k+1} - \underline{v}_l^k)^2 \right] \leq \chi^v, \quad (18d)$$

where χ^s , χ^t , χ^u , and χ^v are small positive constants. The MO, who may be the community manager or an authorized representative of all agents, will collect all traded energy and line prices to check if the algorithm converges, and send a termination signal to all after convergence.

B. Alternating Update Algorithm for Local Optimization Problem

Since above local energy update procedure is still an optimization problem, which is usually solved by professional solvers (e.g., CPLEX and Gurobi), it cannot be implemented by SC, and the operating speed is not fast enough for deploying on a large-scale network or in a short time. In this section, as developed in Appendix A, we propose a primal-dual variables AUA with analytical form to solve the local optimization problem to obtain $\mathbf{E}^{k+1}$. An illustration of the algorithm is shown by Fig. 3. The algorithm can greatly improve the computational efficiency and can be run automatically by SC. The convergence of AUA is analyzed and discussed in Appendix B.

Alternating Update for Seller: In this process, we omit the index k and iterations are indexed by s . For a seller, the traded energy amount are updated as

$$E_n^{s+1} = \max \left(\min \left(\mathcal{V}_n^k + \frac{\tilde{\delta}_n^s}{2a_n + \frac{\rho}{N_B}}, \bar{E}_n \right), E_n \right) \quad (19a)$$

$$\mathbf{E}_n^{s+1} = \left[\mathcal{W}_n^k + \frac{\Delta \mu_n^s - 2a_n E_n^{s+1}}{\rho} \times \mathbf{1}^T \right]^+ \quad (19b)$$

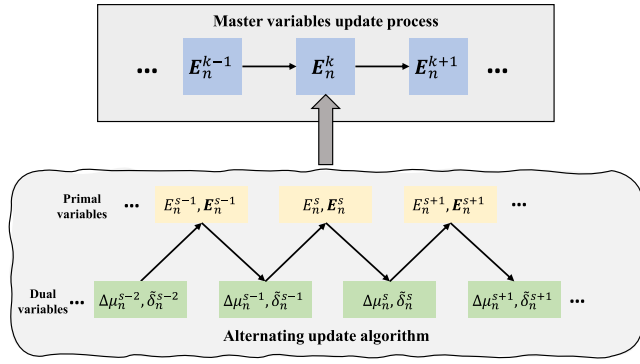


Fig. 3. An illustration of the alternating update algorithm.

Here, $\mathcal{W}_n^k = \{\mathcal{W}_{n1}^k, \dots, \mathcal{W}_{nm}^k\}$, and

$$\mathcal{V}_n^k = \left(\tilde{\lambda}_n^k + \rho \tilde{F}_n^k + \sum_{l \in \mathcal{L}} \varphi_{ij}^l \Delta \tilde{v}_l^k - b_n \right) / \left(2a_n + \frac{\rho}{N_B} \right) \quad (20a)$$

$$\mathcal{W}_{nm}^k = \left(\lambda_{nm}^k + \rho F_{nm}^k - b_n + \sum_{l \in \mathcal{L}} \varphi_{ij}^l \Delta v_l^k \right) / \rho \quad (20b)$$

where $\tilde{\cdot}$ denotes average value. The dual variables $\{\Delta \mu_n, \Delta \tilde{\delta}_n\}$ are updated as

$$\Delta \mu_n^{s+1} = \left(2a_n + \frac{\rho}{N_B} \right) (E_n^{s+1} - \mathcal{V}_n^k) - \tilde{\delta}_n^s \quad (21a)$$

$$\tilde{\delta}_n^{s+1} = \rho (\tilde{E}_n^{s+1} - \tilde{\mathcal{W}}_{nm}^k) - \Delta \mu_n^{s+1} + 2a_n E_n^{s+1} \quad (21b)$$

The primal variables $\{E_n^{s+1}, \mathbf{E}_n^{s+1}\}$ and dual variables $\{\Delta \mu_n^{s+1}, \Delta \tilde{\delta}_n^{s+1}\}$ will be updated in an alternating way until $|E_n^{s+1} - \sum_{m \in \Omega_B} E_{nm}^{s+1}| < \epsilon$ is satisfied.

Alternating Update for Buyer: For a buyer, the traded energy amount are updated as

$$E_m^{s+1} = \max \left(\min \left(\mathcal{V}_m^k + \frac{\tilde{\delta}_m^s}{2Q_m + \frac{\rho}{N_S}}, \bar{E}_m \right), \underline{E}_m \right) \quad (22a)$$

$$\mathbf{E}_m^{s+1} = \left[\mathcal{W}_m^k + \frac{\Delta \mu_m^s - 2Q_m E_m^{s+1}}{\rho} \times \mathbf{1}^T \right]^+ \quad (22b)$$

Here, $\mathcal{W}_m^k = \{\mathcal{W}_{m1}^k, \dots, \mathcal{W}_{mm}^k\}$, and

$$\mathcal{V}_m^k = \left(-\tilde{\lambda}_m^k + \rho \tilde{F}_m^k - \sum_{l \in \mathcal{L}} \varphi_{ji}^l \Delta \tilde{v}_l^k + \varpi_m \right) / \left(2Q_m + \frac{\rho}{N_S} \right) \quad (23a)$$

$$\mathcal{W}_{mm}^k = \left(-\lambda_{mm}^k + \rho F_{mm}^k + \varpi_m - \sum_{l \in \mathcal{L}} \varphi_{ji}^l \Delta v_l^k \right) / \rho \quad (23b)$$

The dual variables $\{\Delta \mu_n, \Delta \tilde{\delta}_n\}$ are updated as

$$\Delta \mu_m^{s+1} = \left(2Q_m + \frac{\rho}{N_S} \right) (E_m^{s+1} - \mathcal{V}_m^k) - \Delta \tilde{\delta}_m^s \quad (24a)$$

$$\tilde{\delta}_m^{s+1} = \rho (\tilde{E}_m^{s+1} - \tilde{\mathcal{W}}_{mm}^k) - \Delta \mu_m^{s+1} + 2Q_m E_m^{s+1} \quad (24b)$$

The primal variables $\{E_m^{s+1}, \mathbf{E}_m^{s+1}\}$ and dual variables $\{\Delta \mu_m^{s+1}, \Delta \tilde{\delta}_m^{s+1}\}$ will be updated in an alternating way until $|E_m^{s+1} - \sum_{n \in \Omega_S} E_{mn}^{s+1}| < \epsilon$ is satisfied.

In conclusion, by converting the local optimization problem into the closed-form AUA, the computational efficiency can be improved. Our final proposed negotiation mechanism can simultaneously enjoy the less iterations of ADMM and high computational speed of analytical computation.

C. PoSo-Based Consensus Mechanism for P2P Energy Trading

The proposed negotiation mechanism in this paper is based on the LMP mechanism, whose main shortcoming is that the properties of “incentive compatibility” and “market efficiency” cannot be satisfied. Previous works on incentive-compatible P2P energy trading mechanisms mainly utilize the VCG mechanism [37], [38], [39], but it cannot satisfy the revenue adequacy property. In this section, we propose a PoSo-based consensus mechanism as an efficient way to prevent untruthful behavior of agents. PoSo [31] is inspired by the PoW mechanism. Similar to PoW, the PoSo mechanism requires participants to solve an optimization problem, and the results are easy to verify. But unlike PoW, the problem in PoSo is meaningful and the result is valuable.

Looking back at the flow chart of PoSo, we find that a delegation is selected from participants and only the temporary leader is responsible for solving the problem and obtaining an optimal solution, while the followers in the delegation verify the solution to decide whether to accept it. However, this paradigm is not suitable for P2P energy trading because agents are unlikely to share their individual problem that contains private information. In this paper, we design a revised version of the PoSo mechanism for our P2P trading mechanism. Compared to the previous work, the main difference is that we do not require delegation. In our consensus mechanism, agents do not need to pass all information to a central leader to solve the optimization problem. The agents parallelly solve individual optimization problem, and send the results to neighboring agents for verification by the deployed smart contract.

Thus, compared to PoSo [31], the main advantage of our consensus mechanism is that the private information can be well protected. However, in order to realize the privacy protection, the computational efficiency may suffer because everyone needs to solve an optimization problem and each neighboring agent needs to verify the result. PoSo has good computational efficiency, but may suffer from the loss of privacy and single point of failure.

To be specific, since the optimization problem of each agent is convex, a solution $\mathbf{E}^*$ is optimal if and only if the KKT condition holds at $\mathbf{E}^*$. Apparently, verifying whether a given $\mathbf{E}^*$ satisfies the optimality conditions is much easier than finding it. The KKT conditions for local updates are (26) and (30). However, checking all of these equations involves a very large communication overhead. Instead of checking all equations, we choose to check the cumulative result (27) and (31). If we take the seller as an example, the cumulative

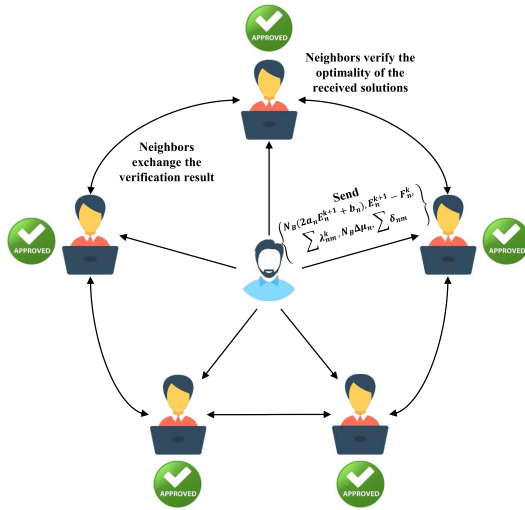


Fig. 4. An illustration of the consensus process of our mechanism.

KKT condition is

$$\begin{aligned}
 & N_B(2a_n E_n + b_n) \\
 &= \sum_{m \in \Omega_B} \lambda_{nm}^k - \rho(E_n - F_n^k) \\
 &+ \sum_{m \in \Omega_B} \sum_{l \in \mathcal{L}} \varphi_{ij}^l (\underline{v}_l^k - \bar{v}_l^k) + N_B \Delta \mu_n + \sum_{m \in \Omega_B} \delta_{nm} \quad (25)
 \end{aligned}$$

After each update, agent n will send $\{N_B(2a_n E_n^{k+1} + b_n), E_n^{k+1} - F_n^k, \sum \lambda_{nm}^k, N_B \Delta \mu_n, \sum \delta_{nm}\}$ to all neighboring agents to check. The term $\sum_{m \in \Omega_B} \sum_{l \in \mathcal{L}} \varphi_{ij}^l (\underline{v}_l^k - \bar{v}_l^k)$ can be obtained from LM. If equation (25) is satisfied, the optimality of the received solution is met, and they will exchange the verification results until all neighboring agents are approved. If the verification fails, agent n will be subject to a substantial fine, which can be formulated as a penalty function added to the objective function. The fine will be divided equally among all neighboring agents. This consensus mechanism can encourage agents to honestly participate in the market, achieving an incentive-compatible market.

It is worth noting that, the verification equation includes the previous broadcast solution $\{\lambda_{nm}^k, F_n^k\}$, which means that, if a malicious agent constructs and broadcasts an inauthentic solution $\{\lambda_{nm}^k, F_n^k\}$ at a certain iteration, it does not pass the next verification unless it continuously uses the fake results to solve the local problem, which may cause economic losses instead. The consensus mechanism is displayed in Fig. 4.

D. Implementation of P2P Energy Trading

We have developed a decentralized negotiation algorithm among agents, but this only addresses part of the problems in the design of P2P energy trading mechanism, and there are still obvious deficiencies:

- Information exchanged among agents is not guaranteed against cyber-attack or tampering by adversaries.
- Individual agent cannot verify that they are being paid/billed as the agreed results.

In this paper, we propose to utilize blockchain technology to create a P2P energy trading platform that takes full advantage of the decentralized structure of the problem, while ensuring operational feasibility and fair payment to all agents. Our algorithm is a natural fit for implementation on a blockchain, as it guarantees convergence but has a less computationally expensive aggregation step [40]. In this work, we utilize SCs to implement the algorithm execution, and the steps are detailed as below.

Step 1: Each seller and buyer has local SC SC_{LO} to run the AUA (19)-(24) to obtain optimal solutions $\{E_n^{k+1}, E_m^{k+1}\}$, and update the energy prices λ_n^{k+1} , which will be broadcast to neighboring agents.

Step 2: Each seller and buyer will verify the solutions $\{E_n, E_m\}$ according to the PoSo-based consensus mechanism (25).

Step 3: Each LM will collect all agents' energy information to execute the power flows calculation (10) and line prices update (17) using SC SC_{LM} .

Step 4: Above steps will iteratively run until all agents reach a consensus on the trading results. The convergence checking procedure (18) is conducted by SC SC_{CO} on MO. If the stopping criteria are all satisfied, it will send a termination signal to all agents.

Step 5: Finally, after convergence, the transactions among sellers and buyers will be packed and stored on the blockchain by SC SC_{TR} , for the aim of verifying that they are being paid/billed as the agreed results in actual operation.

A brief pseudo-code of the SC implementation is presented in Algorithm 1.

IV. NUMERICAL RESULTS

This section presents numerical results to evaluate the performance of our proposed P2P energy trading mechanism using different case studies. For illustration and discussion, a 13-node distribution network derived from IEEE 13 buses system with 4 sellers and 3 buyers is considered as in [6]. Since we want to investigate how the congestion of lines affects and changes the final transactions, the presented test system is transformed into a meshed configuration by adding two additional lines (connecting nodes 2 to 8 and 6 to 10). The sellers are conventional generators and the buyers are households with controllable applications. Compared to the work [6], we use a different high-efficiency market clearing algorithm that can greatly reduce the number of iterations for convergence while maintaining good computational efficiency.

As shown in Fig. 5, each agent is connected to one bus, with sellers on buses 2, 5, 8, and 10, and buyers on buses 3, 4, and 9. Bus 1 is the reference bus. The solid lines indicate the physical electrical connections, and all agents can communicate with each other through a connected network. The parameters of sellers and buyers are listed in Tab. I, which are randomly chosen for testing. The susceptance of each branch is set to 10s, and all stopping criteria χ are set to 10^{-4} . Simulations are run on a computer with an Intel Core i7-10700 processor at 2.90 GHz using 32 GB of RAM.

Algorithm 1: Pseudo-Code of Smart Contracts Implementation for P2P Energy Trading Platform

SC_{LO}:
Input : $\{\lambda_n^k, \mathbf{E}_n^k, \mathbf{E}_m^k\}_{\forall n \in \Omega_S, m \in \Omega_B}, \{\mathbf{v}_l^k\}_{\forall l \in \mathcal{L}}$
Output: $\{\lambda_n^{k+1}, \mathbf{E}_n^{k+1}, \mathbf{E}_m^{k+1}\}_{\forall n \in \Omega_S, m \in \Omega_B}$
if caller == seller **then**

 while $|E_n - \sum_{m \in \Omega_B} E_{nm}| > \epsilon$ **do**

 $\{E_n^{s+1}, E_{nm}^{s+1}\} \leftarrow (19);$

 end
else

 while $|E_m - \sum_{n \in \Omega_S} E_{mn}| > \epsilon$ **do**

 $\{E_m^{s+1}, E_{mn}^{s+1}\} \leftarrow (22);$

 end
end
 $\lambda_n^{k+1} \leftarrow (15);$

SC_{LM}:
Input : $\{\mathbf{E}_n^{k+1}\}_{\forall n \in \Omega_S}, \{\mathbf{v}_l^k\}_{\forall l \in \mathcal{L}}$
Output: $\{\mathbf{v}_l^{k+1}\}_{\forall l \in \mathcal{L}}$
for $\forall l \in \mathcal{L}$ **do**

 $P_l^{k+1} \leftarrow (10);$

 $\{\bar{\mathbf{v}}_l^{k+1}, \underline{\mathbf{v}}_l^{k+1}\} \leftarrow (17);$
end

SC_{CO}:
Input : $\{\mathbf{E}_n^{k,k+1}, \mathbf{E}_m^{k,k+1}\}_{\forall n \in \Omega_S, m \in \Omega_B}, \{\mathbf{v}_l^{k,k+1}\}_{\forall l \in \mathcal{L}}$
Output: Terminal signal

 $\{S^{k+1}, T^{k+1}, V^{k+1}\} \leftarrow (18);$
if Satisfy stopping criteria **then**

 Terminal signal $\leftarrow$ True;

else

 Terminal signal $\leftarrow$ False;

end

SC_{TR}:
if caller == MO **then**

Store transactions into blockchain;

else

Reject the request;

end

We use software Ganache to establish a private Ethereum Homestead blockchain test network, and Python/Gurobi to run the local optimization problems. Remote procedure calls through Web3.py/HTTP allow the Python scripts to communicate with the smart contracts. For development of smart contracts, solidity language is used, which is a special language for smart contracts on Ethereum.

A. Convergence Performance

In this case study, we will show the convergence performance of our designed algorithm. The tuning parameters ρ and ϕ are both set to 1. Fig. 6 displays the convergence process of the algorithm with and without congested line, Tab. II–III shows the final transactions among agents, and

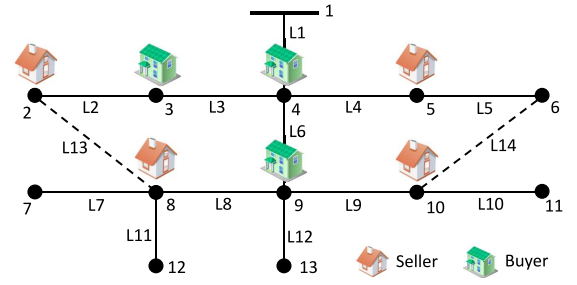
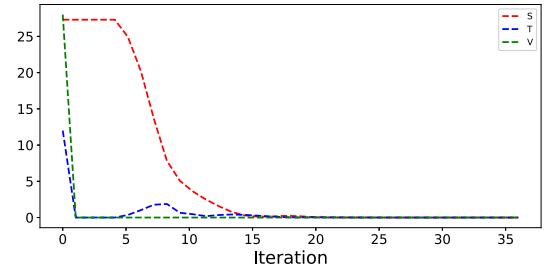


Fig. 5. Meshed Test system schematic.

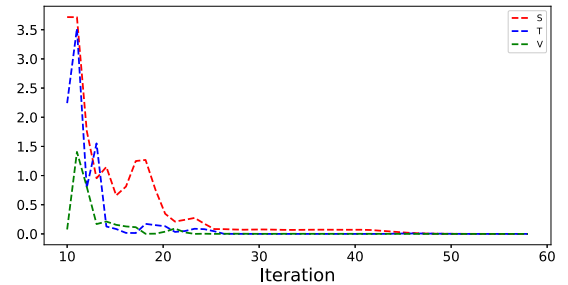
TABLE I
SELLERS' AND BUYERS' PARAMETERS OF SIMPLE CASE STUDY

Seller	Bus	a_n [\$/kW ²]	b_n [\$/kW]	$\underline{E}_n$ [kW]	$\bar{E}_n$ [kW]
S1	2	0.084	5.21	0	6.77
S2	5	0.075	6.35	0	5.17
S3	8	0.051	7.98	0	4.42
S4	10	0.079	8.25	0	2.57

Buyer	Bus	ϱ_m [\$/kW ²]	ϖ_m [\$/kW]	$\underline{E}_m$ [kW]	$\bar{E}_m$ [kW]
B1	3	0.072	8.21	2.35	6.51
B2	4	0.084	11.21	1.46	5.42
B3	9	0.045	13.24	2.16	6.12



(a) No congested line



(b) Congestion in line 2

Fig. 6. Convergence process of the proposed algorithm with and without congestion line.

Tab. IV shows the power flow in each line with and without congestion. In the final transactions, the price consists of two parts: energy price λ_{nm} and line price $\sum_{l \in \mathcal{L}} \phi_{ij}^l (\bar{v}_l - \underline{v}_l)$.

The maximum line capacity P_l^{max} is set to 10 for all lines in the case where there is no overloaded line. It can be seen from Fig. 6(a) that all trading between sellers and buyers converges after about 37 iterations. The profile of V quickly drops to zero and remains at this value as no lines are congested, which also

TABLE II
FINAL TRANSACTIONS WITHOUT CONGESTED LINE

	B1	B2	B3
S1	1.47kW/8.18\$/kW	2.21kW/8.18\$/kW	3.09kW/8.18\$/kW
S2	0.57kW/8.18\$/kW	2.18kW/8.18\$/kW	2.42kW/8.18\$/kW
S3	0.32kW/8.18\$/kW	1.03kW/8.18\$/kW	0.62kW/8.18\$/kW
S4	0kW/8.22\$/kW	0kW/8.20\$/kW	0kW/8.22\$/kW

TABLE III
FINAL TRANSACTIONS WITH CONGESTION IN LINE 2

	B1	B2	B3
S1	0.68kW/12.04\$/kW	1.70kW/10.47\$/kW	2.76kW/9.21\$/kW
S2	0.40kW/12.04\$/kW	2.17kW/10.47\$/kW	2.59kW/9.21\$/kW
S3	0kW/12.07\$/kW	0kW/10.48\$/kW	0kW/9.24\$/kW
S4	1.28kW/12.04\$/kW	0.52kW/10.47\$/kW	0.77kW/9.21\$/kW

TABLE IV
POWER FLOW (kW) IN EACH LINE WITHOUT CONGESTION
(CASE 1) AND WITH CONGESTION IN LINE 2 (CASE 2)

	L2	L3	L4	L5	L6	L8	L9	L13	L14
Case 1	4.38	2.02	-3.78	1.39	0.37	4.37	-1.39	2.39	1.39
Case 2	3.00	0.64	-4.36	0.81	0.62	2.13	-3.38	2.13	0.81

results in the final settlement prices being the same for all non-zero transactions (8.18\$/kW). Since the parameters of S4 are significantly higher than the others ($a_n = 0.079$, $b_n = 8.25$), buyers are unwilling to trade with S4.

However, when there is a congested line, the seller must pay a line utilization charge for transferring power to a particular buyer, resulting in a higher traded price for that buyer and a lower willingness to trade. In another case study, we lower the line capacity in L2 ($P_{l=2}^{max} = 3$) to make it congested and see how the final transactions will change. The line price $\bar{v}_{l=2}$ will increase from 0 to 3.77. Fig. 6 (b) shows that the number of iterations for convergence increases from 37 to 58. The comparison of the trading results in Tab. II–III and the power flows in Tab. IV shows that the energy traded between S1 and B1/B2 is significantly lower, and the prices are higher because these trades require the utilization of the congested line 2 for energy transmission, which imposes additional cost on the trade. Therefore, buyers prefer to purchase energy from other sellers instead of S1 to avoid using the congested line. Another interesting aspect is that when L2 is congested, buyers do not trade with S3 but buy energy from S4, since the sum of energy purchase cost and line usage fee of S3 is higher than that of S4. In addition, the transmission capability of the grid is reduced due to the limited line capacity and the line usage charge. From Tab. II–IV, it can be seen that the total trading volume and transmitted power in the network are decreasing. Thus, our mechanism can motivate agents to adjust their own traded power to reduce congestion and power loss in the network.

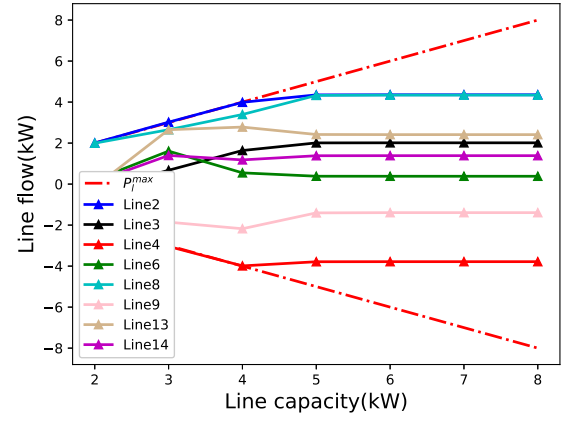


Fig. 7. Power flows in different line for different line limits.

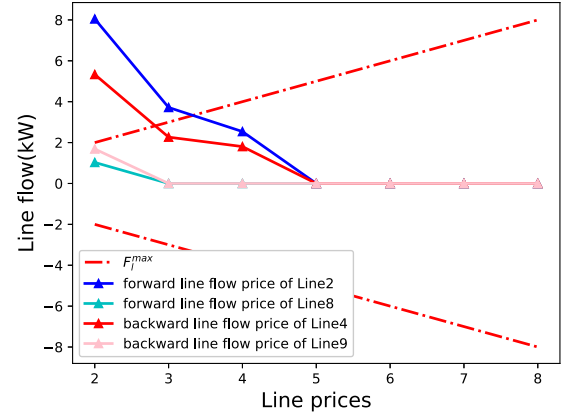


Fig. 8. Line prices in different lines for different line limits.

B. Line Congestion Management Performance

This section examines the impact of line capacity on power flow and line prices. The results are displayed in Fig. 7–8, where only nonzero results are shown. The maximum line capacity for these lines ranges from 2 to 8 kW. It can be seen that the power flows on these lines are always within the maximum line capacity, indicating that the proposed algorithm can meet the line flow constraints in the P2P power network. If there is a congested line in the network, agents will avoid trading through that line since they will have to pay additional network fees. For example, if the line capacity decreases, lines 2 and 8 will be congested, and the forward line prices of lines 2 and 8 will increase, which will cause agents to avoid transferring energy through these lines, and then the line flows will decrease. Similar results are also shown for lines 4 and 9, where the backward line prices will increase to induce agents to adjust the energy transfer.

C. Scalability

In practice, the P2P energy trading network is very large with a large number of agents, and transaction volumes will be significant. The computation time and number of iterations are two key factors that measure the scalability of the mechanism. Fig. 10 shows the impact of the number of agents (between 70 and 420) on these two factors. In this simulation, we do not

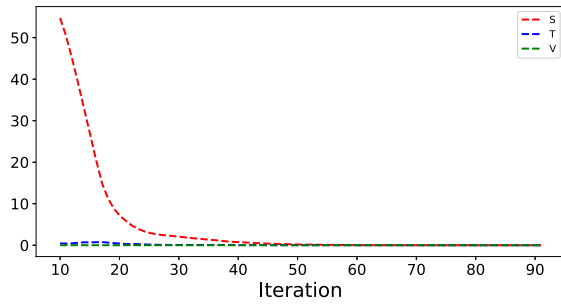


Fig. 9. The convergence performance of the proposed mechanism under IEEE 118-node test feeder.

consider the time consumed by the smart contracts because we focus on the performance of the algorithm. The computation time is approximately equal the total time divided by the number of agents, since all agents have close computational resources. It can be seen that both the computation time and the number of iterations increase with the size of the network. The scalability performance is acceptable (under 5s for over 400 agents), which proves that our proposed mechanism is practical for large-scale markets.

Another simulation runs on IEEE 118-node system to test the validity and scalability of our method in a large system. In the simulation, the parameters of the agents are chosen uniformly in a reasonable range. The convergence performance is shown in Fig. 9. From the result, it can be seen that the iterations for convergence are around 90 and the performance is still good even for a larger system.

D. Comparative Results About High-Efficiency and Incentive-Compatible

In terms of computational efficiency, we compare our method with the consensus+innovation method [8] and the primal-dual gradient method [6], both of which update the derivatives of variables rather than solving an optimization problem. It can be seen that firstly, compared with the CADMM approach, the computation time of CADMM + AUA is greatly reduced, which is only about 1/5 of that of the CADMM approach. Our proposed algorithm can greatly increase the computational efficiency, but still has the low iterations of CADMM. Compared with C+I and primal-dual gradient methods, the computation time of gradient-based methods is lower than our method, but the iterations for convergence are much higher than our method (about two to three times), resulting in extremely high communication overhead. In addition, the iterations for convergence are very sensitive to the chosen tuning parameters. In summary, our proposed method (CADMM + AUA) has the lowest iterations, which can greatly reduce the communication burden, and has good computational efficiency compared to the gradient-based method (close to the C+I method), which is fast enough to be used in practice.

For the “Incentive-Compatible” property, we propose a modified PoSo in this paper, which is an efficient consensus mechanism to resist agents’ untruthful behavior. We added an example as shown in Fig. 12 to demonstrate it by comparing

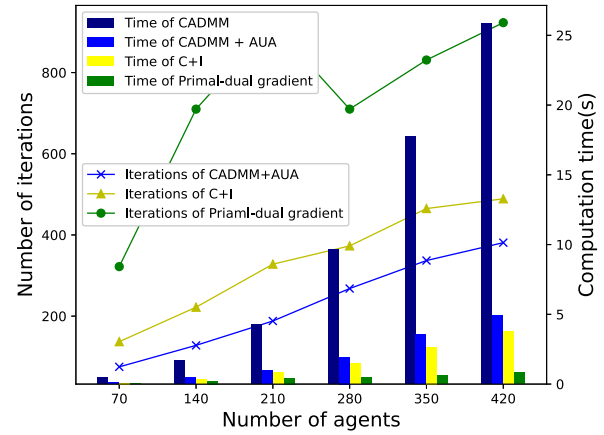


Fig. 10. Impact of number of agents on computation time and number of iterations for convergence.

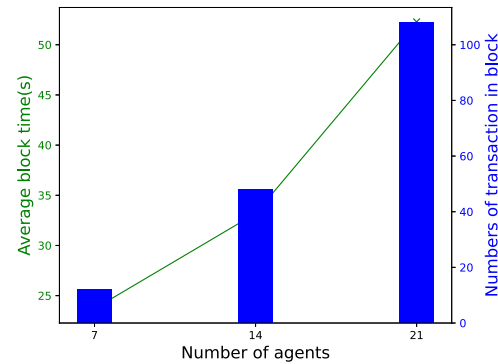


Fig. 11. The average block time and number of transactions in block with the increase of agents.

it with LMP-based P2P energy trading without the consensus mechanism. We continue to use the 13-node network and agent parameters as in Section IV. The line capacity is set to 10. From the Table II, we see that the marginal price is determined by S3 since S1 and S2 both reach the maximum capacity. Without the consensus mechanism and smart contract, S3 can adopt a strategy to achieve higher profit. It is assumed that S3 can get all the information from each agent and calculate the optimal solutions of the market. S3 can know the final transaction prices, which are the marginal price of the market (8.18 \$/kW). In the negotiation process, S3 will then deliberately increase the price it offered to buyers. After a sufficient number of iterations (100), S3 will adjust the trade amount received from the buyers to complete the negotiation. The welfare of social and individual agents under the unveracious price offered by S3 with and without consensus mechanism is shown in Fig. 12. It can be seen that as the price offered by S3 is increased, the welfare of S3 gradually increases until it reaches the maximum value. If S3 continues to raise the price, buyers will switch to buying more energy from S4, resulting in a lower profit for S3. Social welfare decreases as the price of S3 increases because buyers have to buy energy from the uneconomic S4. Therefore, without the consensus mechanism, S3 can maximize its own interests by harming the social welfare, in other words, the incentive compatibility cannot be

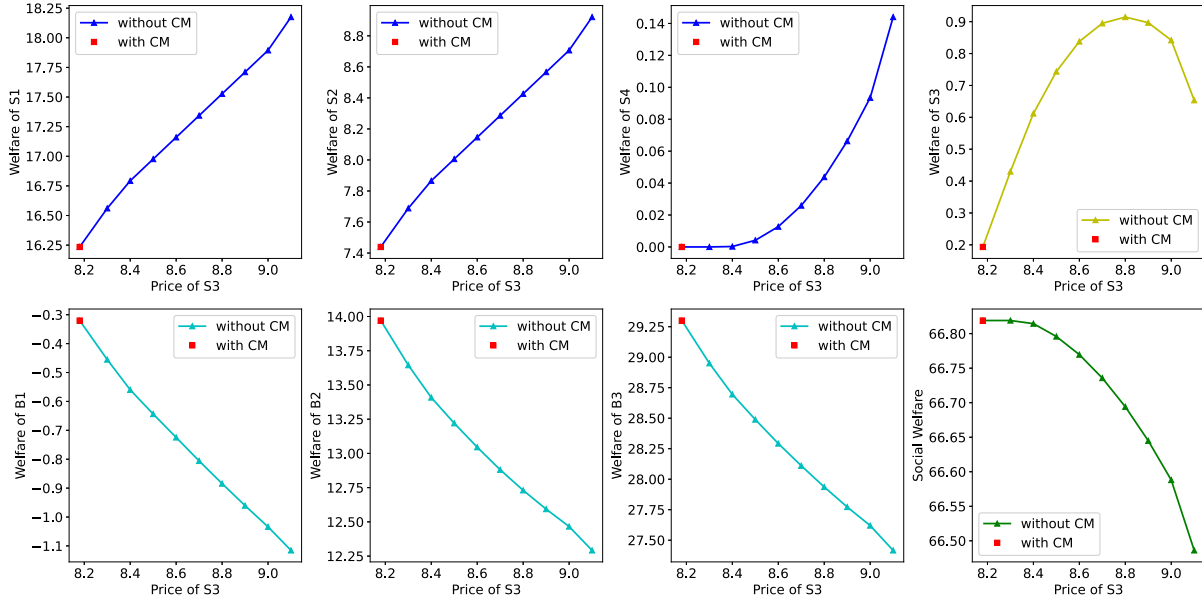


Fig. 12. The welfare of social and each agent under the inveracious price offered by S3 with and without consensus mechanism (CM).

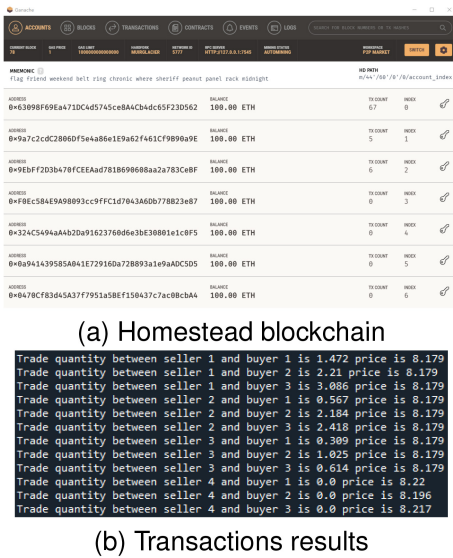


Fig. 13. P2P energy trading homestead blockchain and transaction results stored in blockchain.

satisfied. However, under the consensus mechanism proposed in our paper, the offered inveracious price will not pass the verification by others, and the smart contract also forces S3 to follow the rules.

E. Blockchain-Based P2P Energy Trading Platform

We use Ganache to set up a private Ethereum Homestead blockchain named “P2P Market”, as shown in Fig. 13. Ganache is a personal blockchain for Ethereum development, which can be used to deploy contracts, develop your applications, and run tests. Addresses are assigned to MO, each LM and each agent. The gas fee is set very low to facilitate simulation. Smart contract SC_{LO} is deployed on each agent to run the AUA and determine the bid quantities and prices to be

shared with neighbors, LMs, and MO. Each agent executes the consensus mechanism to verify the optimality of the solutions received from neighboring agents. SC_{LM} on each LM updates power flows and line prices, and return to particular agents. SC_{CO} on MO controls the convergence process, and SC_{TR} provides functions to store and get trading results, which can be checked by everyone in the network.

Fig. 11 shows the performance of the P2P energy trading platform. It can be seen that, with the increase of agents, the average block time and number of transactions in block will both increase accordingly, since the number of iterations for convergence and the time needed to run a SC will increase by the number of agents.

V. CONCLUSION, LIMITATION AND FUTURE PERSPECTIVE

A. Conclusion

P2P energy trading mechanisms are considered as an evolution of the power network, but the “P2P energy trading trilemma” remains a great challenge. Moreover, LMP-based market mechanisms cannot satisfy the incentive compatibility property. In this paper, a novel high-efficiency and incentive-compatible P2P negotiation mechanism is proposed. The local optimization problem is transformed into a closed-form AUA so that it can be implemented using SC and the computational efficiency is greatly improved. In addition, we design a PoSo-based consensus mechanism for P2P energy trading. The optimality of the solutions is verified by neighboring agents. Various case studies are conducted in simulations to demonstrate the validity, reliability, and scalability of our mechanism. We also provide comparative computational efficiency results with existing P2P mechanisms, showing that compared to gradient-based methods, the iterations for convergence are much lower and the computational speed is acceptable and good enough for practical application. Moreover, we prove that incentive-compatibility can be realized under our proposed

consensus mechanism. Finally, a blockchain-based P2P energy trading platform is also built for SC operations.

B. Limitation and Future Perspective

Some improvements and future works are needed to overcome the limitations of our work and adapt to the actual system. First, we may consider use the more practical AC power flow model, which can be exactly relaxed by second order cone programming (SOCP) method [41]. We still consider introducing a SO into the market, who acts as a single agent and is responsible for solving the ACOF problem; Second, in practice, prosumers may be equipped with energy storage systems, and binary integer variables are usually introduced for modeling. In our future work, we will focus on combining l_p -Box ADMM [42] with our method to design negotiation mechanism for P2P energy trading with integer variables; Third, for the P2P energy trading consensus mechanism, the difficulty in practical application lies in the high timeliness requirement of the market. The idea of improvement is to randomly choose time slots and agents to execute consensus verification, which can effectively reduce the time spent while ensuring the correct transaction results. Future works also include the sparsity of the communication matrix using network segmentation or blockchain sharding technology, and developing a privacy-preserving mechanism for P2P energy trading based on privacy computing techniques.

APPENDIX A

ALTERNATING UPDATE ALGORITHM FOR LOCAL OPTIMIZATION PROBLEM

A. Alternating Update Algorithm for Sellers

For a seller, since the local optimization problem (13) is convex, it can be equivalently solved by its KKT conditions. Here, we replace the variable E_n by using the equality constraint $E_n = \sum_{m \in \Omega_B} E_{nm}$, and the constraint $E_{nm} \leq E_n$ can be removed. The KKT conditions are a nonlinear problem with primal variables E_n and dual variables $\{\underline{\mu}_n, \bar{\mu}_n\}$, δ_{nm} for constraints $\underline{E}_n \leq E_n \leq \bar{E}_n$ and $0 \leq E_{nm}$. In this process, iterations are indexed by s .

The KKT condition of (14) for E_{nm} is formulated as

$$\begin{aligned} \partial C_n(E_n) - \lambda_{nm}^k + \rho(E_{nm} - F_{nm}^k) \\ + \sum_{l \in \mathcal{L}} \varphi_{ij}^l(\bar{v}_l^k - \underline{v}_l^k) - \underline{\mu}_n + \bar{\mu}_n - \delta_{nm} = 0. \end{aligned} \quad (26)$$

Here F_{nm} is the consensus variable equals to $\frac{E_{nm} + E_{mn}}{2}$. Summing up (26) for all $m \in \Omega_B$, we have

$$\begin{aligned} N_B(2a_n E_n + b_n) = \sum_{m \in \Omega_B} \lambda_{nm}^k - \rho(E_n - F_n^k) \\ + \sum_{m \in \Omega_B} \sum_{l \in \mathcal{L}} \varphi_{ij}^l(\underline{v}_l^k - \bar{v}_l^k) + N_B(\underline{\mu}_n - \bar{\mu}_n) \\ + \sum_{m \in \Omega_B} \delta_{nm} \end{aligned} \quad (27)$$

For simplicity, $\mu - \bar{\mu}$ and $\underline{v}_l - \bar{v}_l$ are denoted by $\Delta\mu$ and Δv_l , respectively. Given the dual variables results from last

iteration s , E_n^{s+1} is updated as

$$E_n^{s+1} = \mathcal{V}_n^k + \frac{\Delta\mu_n^s + \tilde{\delta}_n^s}{2a_n + \frac{\rho}{N_B}}. \quad (28)$$

According to the complementary condition, E_n^{s+1} can be updated in another equivalent form as (19a). According to (26), the trading amount to buyer $m \in \Omega_B$ is updated as

$$E_{nm}^{s+1} = \mathcal{W}_{nm}^k + \frac{\Delta\mu_n^s + \delta_{nm}^s - 2a_n E_n^{s+1}}{\rho}, \quad (29)$$

Similarly, we can update E_{nm}^{s+1} in another form based on complementary condition as (19b). From (28), the dual variables $\Delta\mu_n^{s+1}$ is updated as (21a). From (29), the average dual variable $\Delta\tilde{\delta}_n^{s+1}$ is updated as (21b).

B. Alternating Update Algorithm for Buyers

Similarly, for a buyer, the KKT conditions of (16) for a trade with buyer m is formulated as

$$\begin{aligned} -\partial U_m(E_m) + \lambda_{nm}^k + \rho(E_{mn} - F_{mn}^k) \\ - \sum_{l \in \mathcal{L}} \varphi_{ji}^l(\bar{v}_l^k - \underline{v}_l^k) - \underline{\mu}_m + \bar{\mu}_m - \delta_{mn} = 0. \end{aligned} \quad (30)$$

Summing up (30) for all $n \in \Omega_S$, we have

$$\begin{aligned} N_S(2Q_m E_m - \varpi_m) = - \sum_{n \in \Omega_S} \lambda_{nm}^k - \rho(E_m - F_m^k) \\ - \sum_{n \in \Omega_S} \sum_{l \in \mathcal{L}} \varphi_{ji}^l(\underline{v}_l^k - \bar{v}_l^k) \\ + N_S(\underline{\mu}_m - \bar{\mu}_m) + \sum_{n \in \Omega_S} \delta_{mn}. \end{aligned} \quad (31)$$

Given the dual variables results from last iteration s , E_m^{s+1} is updated as

$$E_m^{s+1} = \mathcal{V}_m^k + \frac{\Delta\mu_m^s + \tilde{\delta}_m^s}{2Q_m + \frac{\rho}{N_S}}. \quad (32)$$

According to the complementary condition, E_m^{s+1} can be updated in another equivalent form as (22a). According to (30), the trading amount to seller $n \in \Omega_S$ is updated as

$$E_{mn}^{s+1} = \mathcal{W}_{mn}^k + \frac{\Delta\mu_m^s + \delta_{mn}^s - 2Q_m E_m^{s+1}}{\rho}, \quad (33)$$

Similarly, we can update E_{mn}^{s+1} in another form based on complementary condition as (22b). From (32), the dual variables $\Delta\mu_m^{s+1}$ is updated as (24a). From (33), the average dual variable $\tilde{\delta}_m^{s+1}$ is updated as (24b).

APPENDIX B

DISCUSSION AND ANALYSIS OF THE CONVERGENCE OF AUA

We take seller n as example to discuss the convergence of AUA. The idea of AUA is inspired by and similar to Gauss-Seidel iterative method. The computation of variables uses other ones that have already been computed. Since the equations (19) and (22) are not linear equations, it is unable to analyze the spectral radius $\rho(M)$ of matrix $M = (D - L)^{-1}U$.

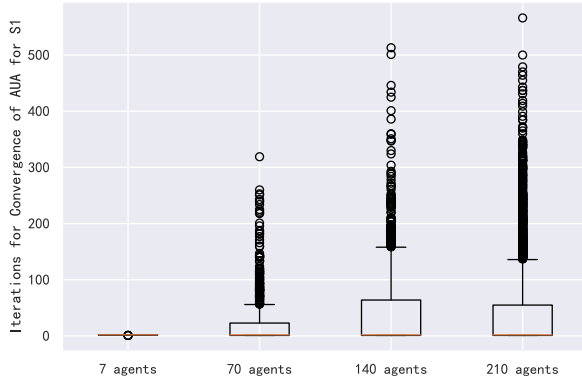


Fig. 14. The iterations for convergence of AUA for seller 1.

But we can still prove the convergence by analyzing the gap $|E_n^{s+1} - \sum_{m \in \Omega_B} E_{nm}^{s+1}|$.

For ease of representation, we first assume that E_n is always within the power boundaries, i.e., $\underline{E}_n < E_n < \bar{E}_n$, which means the dual variable $\Delta\mu_n$ always equals to zero. Thus, in the first iteration $E_n^1 = \mathcal{V}_n^k$ ($\tilde{\delta}_n^0 = 0$), and E_n^1 is calculated as $E_n = [\mathcal{W}_n^k - \frac{2a_n E_n}{\rho} \times \mathbf{1}^T]^+$. Then, we will discuss it in two cases.

- If all $E_{nm}^1 > 0, \forall m \in \Omega_B$, then we have $\tilde{\delta}_n^1 = 0$, and it is easy to verify $E_n^1 = \sum_{m \in \Omega_B} E_{nm}^1$. The AUA converges at one iteration.
- If some $E_{nm}^1 = 0$, then some $\delta_{nm}^1 > 0$ and $\tilde{\delta}_n^1$ is updated by (21b). From $E_{nm} = \mathcal{W}_{nm}^k + \frac{\Delta\mu_n + \delta_{nm} - 2a_n E_n}{\rho}$, we can obtain the gap $\sum_{m \in \Omega_B} E_{nm}^1 - E_n^1$ is $\sum_{m \in \Omega_B} \delta_{nm}^1 / \rho$. In the next iteration, E_n will increase by $\frac{\delta_n^1}{2a_n + \frac{\rho}{N_B}}$, and $\sum_{m \in \Omega_B} E_{nm}$ will at most decrease by $\frac{2a_n N_B}{\rho} \times \frac{\delta_n^1}{2a_n + \frac{\rho}{N_B}}$ (since some E_{nm} may reach the zero boundary). Thus the gap $\sum_{m \in \Omega_B} E_{nm}^2 - E_n^2$ will at most reduce by $\frac{\delta_n^1}{2a_n + \frac{\rho}{N_B}} + \frac{2a_n N_B}{\rho} \times \frac{\delta_n^1}{2a_n + \frac{\rho}{N_B}} = \sum_{m \in \Omega_B} \delta_{nm}^1 / \rho = \sum_{m \in \Omega_B} E_{nm}^1 - E_n^1$. Therefore, in the optimistic case, the AUA usually converges in only two iterations. If not, the gap $\sum_{m \in \Omega_B} E_{nm}^2 - E_n^2$ is updated to be $\sum_{m \in \Omega_B} \delta_{nm}^2 / \rho$, which must have gone down, i.e., $\sum_{m \in \Omega_B} \delta_{nm}^2 / \rho < \sum_{m \in \Omega_B} \delta_{nm}^1 / \rho$.

Thus, in conclusion, using AUA method, the gap $\sum_{m \in \Omega_B} E_{nm} - E_n$ decreases monotonically until it is less than the stopping criteria ϵ .

If E_n reaches the boundaries $\{\underline{E}_n, \bar{E}_n\}$ at the beginning or after some iterations, the dual variable $\Delta\mu_n$ will not be zero, and begin to take effect on the update of E_n . The proof process is similar to above that by adjusting $\Delta\mu_n$ and $\tilde{\delta}_n$, the gap $\sum_{m \in \Omega_B} E_{nm} - E_n$ still decreases monotonically until it is less than stopping criteria ϵ . We also run a simulation to show the iterations for convergence of AUA method for seller 1 in Fig. 14. It can be seen that AUA is sure to converge, and the median is only 1 or 2, which means the algorithm will converge very quickly in most cases.

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